# Percent Tools

Keep this page in your folder. Allowed as a support on classwork, homework and (when your plan allows) tests.

## SAME VALUE, THREE WAYS

*fraction · decimal · percent*

| Fraction | Decimal | Percent |
|---|---|---|
| 1/100 | 0.01 | 1% |
| 1/10 | 0.1 | 10% |
| 1/5 | 0.2 | 20% |
| 1/4 | 0.25 | 25% |
| 1/3 | about 0.333 | about 33.3% |
| 1/2 | 0.5 | 50% |
| 3/4 | 0.75 | 75% |
| 1 | 1 | 100% |
| 3/2 | 1.5 | 150% |

## BUILD ANY PERCENT FROM 10% AND 1%

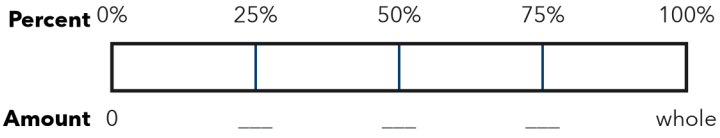
*example: the whole is 80*

| | |
|---|---|
| **10%** | Divide by 10: 10% of 80 = 8 |
| **5%** | Half of 10%: 4 |
| **1%** | Divide by 100: 0.8 |
| **35%** | 3 × 10% + 5% = 24 + 4 = 28 |
| **150%** | 100% + 50% = 80 + 40 = 120 |

## THREE KINDS OF PERCENT PROBLEMS

*find the missing piece*

| | |
|---|---|
| **Find the percent** | part ÷ whole, then × 100. 9 out of 12: 9 ÷ 12 = 0.75 = 75% |
| **Find the part** | percent × whole. 30% of 60: 0.3 × 60 = 18 |
| **Find the whole** | part ÷ percent. 12 is 40%: 12 ÷ 0.4 = 30 (or 40% = 12, 10% = 3, 100% = 30) |




**The whole is always 100%. More than the whole is more than 100%.**